

Note 11: Choi isomorphism, Quantum operation

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11.1 Summary of Progress

1. Proved ChoiJamiokowski isomorphism via linear extension on Kronecker products.
2. Equivalenced three quantum operation definitions and proved AXI $\Rightarrow$ KRA.
3. Analyzed QEC ideal assumptions, hardware leakage noise, leakage Kraus channel and partial-trace block theory.

11.2 Choi - Jamiokowski Isomorphism

- Let $E_{ij} = e_i e_j^\top$, and $M_n = M_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices.
- Denote $\mathcal{L}(U, V) := \text{Hom}_{\text{vect}}(U, V) = \{\text{linear map from } U \text{ to } V\}$.

Theorem 1 (Choi–Jamiokowski Isomorphism) *The following linear maps establish mutually inverse vector space isomorphisms between $\mathcal{L}(M_n, M_m)$ and $M_m \otimes M_n$:*

$$\begin{cases} \mathcal{T} : \mathcal{L}(M_n, M_m) \rightarrow M_m \otimes M_n, & \mathcal{T}(\Phi) = T_\Phi = \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij}) \\ \mathcal{M} : M_m \otimes M_n \rightarrow \mathcal{L}(M_n, M_m), & \Phi_T(X) = (\text{tr} \otimes \text{id}_{M_m})((X^\top \otimes I_m)T) \end{cases}$$

In particular, $\mathcal{M} \circ \mathcal{T} = \text{id}_{\mathcal{L}(M_n, M_m)}$ and $\mathcal{T} \circ \mathcal{M} = \text{id}_{M_m \otimes M_n}$.

Proof We verify that $\mathcal{T} : \mathcal{L}(M_n, M_m) \rightarrow M_m \otimes M_n$ and $\mathcal{M} : M_m \otimes M_n \rightarrow \mathcal{L}(M_n, M_m)$ are mutually inverse linear maps.

(1) Consider elementary tensors $T = A \otimes B$. Substitute into the definition of $\mathcal{M}$:

$$\begin{aligned} \Phi_{A \otimes B}(X) &= (\text{tr} \otimes \text{id}_{M_m}) \left((X^\top \otimes I_m) (A \otimes B) \right) \\ &= (\text{tr} \otimes \text{id}_{M_m}) \left((X^\top A) \otimes B \right) = \text{tr} \left(X^\top A \right) B. \end{aligned}$$

Apply the Choi map $\mathcal{T}$ to this linear operator:

$$\begin{aligned} \mathcal{T}(\Phi_{A \otimes B}) &= \sum_{i,j=1}^n E_{ij} \otimes \Phi_{A \otimes B}(E_{ij}) = \sum_{i,j=1}^n E_{ij} \otimes \text{tr} \left(E_{ij}^\top A \right) B \\ &= \sum_{i,j=1}^n E_{ij} \otimes \text{tr} \left(e_j e_i^\top A \right) B = \sum_{i,j=1}^n E_{ij} \otimes \text{tr} \left(e_i^\top A e_j \right) B. \end{aligned}$$

Denote the matrix entries of A by $a_{ij} = e_i^\top A e_j$, then

$$\mathcal{T}(\Phi_{A \otimes B}) = \sum_{i,j=1}^n a_{ij} E_{ij} \otimes B = \left(\sum_{i,j} a_{ij} E_{ij} \right) \otimes B = A \otimes B.$$

This shows $\mathcal{T} \circ \mathcal{M}(A \otimes B) = A \otimes B$ holds for all pure tensors.

- (2) Extend to arbitrary tensors by linearity. Any tensor $T \in M_m \otimes M_n$ can be expressed as a finite linear combination of elementary tensors:

$$T = \sum_k A_k \otimes B_k.$$

By linearity of $\mathcal{M}$:

$$\Phi_{\sum_k A_k \otimes B_k}(X) = \sum_k \text{tr}(X^\top A_k) B_k.$$

Now apply $\mathcal{T}$:

$$\begin{aligned} \mathcal{T}\left(\Phi_{\sum_k A_k \otimes B_k}\right) &= \sum_{i,j=1}^n E_{ij} \otimes \Phi_{\sum_k A_k \otimes B_k}(E_{ij}) \\ &= \sum_{i,j=1}^n E_{ij} \otimes \sum_k \text{tr}(E_{ij}^\top A_k) B_k \\ &= \sum_k \left(\sum_{i,j=1}^n E_{ij} \otimes \text{tr}(E_{ij}^\top A_k) B_k \right) \\ &= \sum_k A_k \otimes B_k. \end{aligned}$$

Hence $\mathcal{T} \circ \mathcal{M} = \text{id}_{M_m \otimes M_n}$.

- (3) We now prove the reverse identity $\mathcal{M} \circ \mathcal{T} = \text{id}_{\mathcal{L}(M_n, M_m)}$. Take arbitrary $\Phi \in \mathcal{L}(M_n, M_m)$ and let $T_\Phi = \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij})$.

$$\begin{aligned} \mathcal{M}(T_\Phi)(X) &= \mathcal{M}\left(\sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij})\right)(X) \\ &= (\text{tr} \otimes \text{id}_{M_m})\left(\left(X^\top \otimes I_m\right) \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij})\right) \\ &= (\text{tr} \otimes \text{id}_{M_m})\left(\sum_{i,j=1}^n X^\top E_{ij} \otimes \Phi(E_{ij})\right) \\ &= \sum_{i,j=1}^n \text{tr}(E_{ij}^\top X) \Phi(E_{ij}). \end{aligned}$$

For $\text{tr}(E_{ij}^\top X) = \text{tr}(e_j e_i^\top X) = \text{tr}(e_i^\top X e_j) = x_{ij}$ and using linearity of Φ ,

$$\mathcal{M}(T_\Phi)(X) = \sum_{i,j=1}^n x_{ij} \Phi(E_{ij}) = \Phi\left(\sum_{i,j=1}^n x_{ij} E_{ij}\right) = \Phi(X).$$

We conclude $\mathcal{T} \circ \mathcal{M} = \text{id}_{M_m \otimes M_n}$ and $\mathcal{M} \circ \mathcal{T} = \text{id}_{\mathcal{L}(M_n, M_m)}$. Therefore $\mathcal{T}$ and $\mathcal{M}$ are mutually inverse linear isomorphisms. ■

11.3 Quantum operation, Kraus sum representation theorem

Definition 2 (System-Environment Model) Let $\rho \in \mathcal{L}(\mathcal{H}_S)$, $\mathcal{H}_E$ environment with initial state $|e_0\rangle\langle e_0|$. A SEM quantum operation is

$$\mathcal{E}(\rho) = \text{tr}_E \left[U (\rho \otimes |e_0\rangle\langle e_0|) U^\dagger \right],$$

where U is unitary on $\mathcal{H}_S \otimes \mathcal{H}_E$.

Definition 3 (Operator-Sum (Kraus) Representation) A map $\mathcal{E}$ is a **KRA quantum operation** if there exist linear operators $\{E_k\}$ such that

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \quad \sum_k E_k^\dagger E_k \leq I.$$

If $\sum_k E_k^\dagger E_k = I$, $\mathcal{E}$ is trace-preserving (quantum channel).

Definition 4 (Axiomatic Definition) A linear map $\mathcal{E}$ is a **AXI quantum operation** if

1. Trace bound: $0 \leq \text{tr } \mathcal{E}(\rho) \leq \text{tr } \rho$, $\forall \rho \geq 0$;
2. Complete positivity: $\mathcal{I} \otimes \mathcal{E}$ preserves positivity.

Theorem 5 The three formulations of quantum operations (SEM, KRA, AXI) are equivalent.

$$\text{SEM} \iff \text{KRA} \iff \text{AXI}$$

The proof of Kraus sum representation theorem

Lemma 6 If $T_{\mathcal{E}} = \sum_{i,j} E_{ij} \otimes \mathcal{E}(E_{ij})$ the Choi matrix of $\mathcal{E} : \mathcal{L}(\mathcal{H}) \rightarrow \mathcal{L}(\mathcal{H})$, then

$$\left(|\lambda\rangle^\top \otimes I \right) T_{\mathcal{E}} \left(|\overline{\lambda}\rangle \otimes I \right) = \mathcal{E}(|\lambda\rangle\langle\lambda|).$$

Proof Let $|\lambda\rangle = \sum_a \lambda_a |a\rangle$, define

$$|\lambda\rangle^\top \triangleq \sum_a \lambda_a \langle a|, \quad |\overline{\lambda}\rangle \triangleq \sum_a \overline{\lambda_a} \langle a|.$$

We first compute two elementary identities:

$$|\lambda\rangle^\top |i\rangle = \left(\sum_a \lambda_a \langle a| \right) |i\rangle = \sum_a \lambda_a \langle a|i\rangle = \lambda_i,$$

$$\langle j| \overline{|\lambda\rangle} = \langle j| \left(\sum_b \overline{\lambda_b} \langle b| \right) = \sum_b \overline{\lambda_b} \langle j|b\rangle = \overline{\lambda_j}.$$

Let matrix unit $E_{ij} = |i\rangle\langle j|$. Using $(A \otimes B)(C \otimes D) = AC \otimes BD$:

$$\begin{aligned} \left(|\lambda\rangle^\top \otimes I \right) (E_{ij} \otimes \mathcal{E}(E_{ij})) \left(|\overline{\lambda}\rangle \otimes I \right) &= (|\lambda\rangle^\top |i\rangle) \mathcal{E}(E_{ij}) (\langle j| \overline{|\lambda\rangle}) \\ &= \lambda_i \mathcal{E}(|i\rangle\langle j|) \overline{\lambda_j}. \end{aligned}$$

Recall the Choi matrix definition

$$T_{\mathcal{E}} = \sum_{i,j} E_{ij} \otimes \mathcal{E}(E_{ij}).$$

Summing over all i, j and using linearity of $\mathcal{E}$:

$$\begin{aligned} \left(|\lambda\rangle^\top \otimes I\right) T_{\mathcal{E}} \left(|\overline{\lambda}\rangle \otimes I\right) &= \sum_{i,j} \lambda_i \overline{\lambda_j} \mathcal{E}(|i\rangle\langle j|) \\ &= \mathcal{E} \left(\sum_{i,j} \lambda_i \overline{\lambda_j} |i\rangle\langle j| \right). \end{aligned}$$

Recognize the outer product:

$$\sum_{i,j} \lambda_i \overline{\lambda_j} |i\rangle\langle j| = \left(\sum_i \lambda_i |i\rangle \right) \left(\sum_j \lambda_j |j\rangle \right)^\dagger = |\lambda\rangle\langle\lambda|.$$

Substitution yields

$$\left(|\lambda\rangle^\top \otimes I\right) T_{\mathcal{E}} \left(|\overline{\lambda}\rangle \otimes I\right) = \mathcal{E}(|\lambda\rangle\langle\lambda|).$$

□

■

Theorem 7 AXI $\implies$ KRA

Proof Suppose linear map $\mathcal{E}$ satisfies the axiomatic definition (AXI): linearity, trace bound and complete positivity. Define the Choi matrix

$$T_{\mathcal{E}} = (\mathcal{I} \otimes \mathcal{E})(|\Psi\rangle\langle\Psi|) = \sum_{i,j} E_{ij} \otimes \mathcal{E}(E_{ij}).$$

Complete positivity implies $\mathcal{I} \otimes \mathcal{E}$ preserves positivity, so $T_{\mathcal{E}} \succeq 0$. Spectral decomposition yields

$$T_{\mathcal{E}} = \sum_k |s_k\rangle\langle s_k|, \quad |s_k\rangle \in \mathcal{H} \otimes \mathcal{H}.$$

Define linear operators $E_k : \mathcal{H} \rightarrow \mathcal{H}$ by

$$E_k |\psi\rangle \triangleq (|\psi\rangle^\top \otimes I) |s_k\rangle, \quad \forall |\psi\rangle \in \mathcal{H}.$$

Now take an arbitrary pure state $|\lambda\rangle\langle\lambda|$. Direct computation gives

$$\begin{aligned} \sum_k E_k |\lambda\rangle\langle\lambda| E_k^\dagger &= \sum_k (|\lambda\rangle^\top \otimes I) |s_k\rangle \langle s_k| (|\overline{\lambda}\rangle \otimes I) \\ &= (|\lambda\rangle^\top \otimes I) \left(\sum_k |s_k\rangle\langle s_k| \right) (|\overline{\lambda}\rangle \otimes I) \\ &= (|\lambda\rangle^\top \otimes I) T_{\mathcal{E}} (|\overline{\lambda}\rangle \otimes I). \end{aligned}$$

By Lemma 6:

$$(|\lambda\rangle^\top \otimes I) T_{\mathcal{E}} (|\overline{\lambda}\rangle \otimes I) = \mathcal{E}(|\lambda\rangle\langle\lambda|).$$

Thus

$$\sum_k E_k |\lambda\rangle\langle\lambda| E_k^\dagger = \mathcal{E}(|\lambda\rangle\langle\lambda|)$$

holds for all pure states $|\lambda\rangle\langle\lambda|$.

For arbitrary density matrix $\rho \in \mathcal{L}(\mathcal{H})$, perform spectral decomposition

$$\rho = \sum_{\lambda} p_{\lambda} |\lambda\rangle\langle\lambda|, \quad p_{\lambda} \geq 0.$$

By linearity of $\mathcal{E}$:

$$\sum_k E_k \rho E_k^\dagger = \sum_\lambda p_\lambda \sum_k E_k |\lambda\rangle\langle\lambda| E_k^\dagger = \sum_\lambda p_\lambda \mathcal{E}(|\lambda\rangle\langle\lambda|) = \mathcal{E}(\rho).$$

We now verify normalization condition $\sum_k E_k^\dagger E_k \leq I$. From trace bound condition $\text{tr } \mathcal{E}(\sigma) \leq \text{tr } \sigma$ for all positive σ . For any normalized pure state $|\psi\rangle$,

$$\langle\psi| \left(\sum_k E_k^\dagger E_k \right) |\psi\rangle = \text{tr} \left(\sum_k E_k |\psi\rangle\langle\psi| E_k^\dagger \right) = \text{tr } \mathcal{E}(|\psi\rangle\langle\psi|) \leq \text{tr}(|\psi\rangle\langle\psi|) = 1.$$

The inequality holds for all $|\psi\rangle$, hence $\sum_k E_k^\dagger E_k \leq I$.

Therefore $\mathcal{E}$ admits operator-sum representation, so $\mathcal{E}$ is a KRA quantum operation. $\square$ ■

Examples of single-qubit quantum channels

Channel Name	Kraus Operators
Bit flip channel	$E_0 = \sqrt{1-p} \mathbb{I}, \quad E_1 = \sqrt{p} X$
Phase flip channel	$E_0 = \sqrt{1-p} \mathbb{I}, \quad E_1 = \sqrt{p} Z$
Bit-phase flip channel	$E_0 = \sqrt{1-p} \mathbb{I}, \quad E_1 = \sqrt{p} Y$
Depolarizing channel	$E_0 = \sqrt{1 - \frac{3p}{4}} \mathbb{I}, \quad E_1 = \frac{\sqrt{p}}{2} X, \quad E_2 = \frac{\sqrt{p}}{2} Y, \quad E_3 = \frac{\sqrt{p}}{2} Z$
Amplitude damping channel	$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$
Phase damping channel	$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$

Table 11.1: Kraus representations for typical single-qubit quantum channels

Remark: The phase damping channel admits an equivalent Pauli-type Kraus representation via Kraus unitary freedom. Let

$$p = \frac{1}{2}(1 - \sqrt{1-\lambda}),$$

then the alternative Kraus operators are

$$F_0 = \sqrt{1-p} \mathbb{I}, \quad F_1 = \sqrt{p} Z.$$

11.4 Non-Ideal Hardware Noise and the Four Core Assumptions of QEC

- **Assumption 1: Independent errors** Distinct qubit error probabilities factorize:

$$P(e_a, e_b) = P(e_a)P(e_b), \quad a \neq b$$

No correlated propagation or synchronous clustered faults.

- **Assumption 2: Sparse errors** Within one cycle, error weight is bounded by code distance:

$$w(e) \leq t = \left\lfloor \frac{d-1}{2} \right\rfloor$$

Only isolated single-qubit failures exist, no large contiguous fault clusters.

- **Assumption 3: Instantaneous errors** Noise only lasts one stabilizer measurement cycle; qubits are fully refreshed each round, no cross-cycle persistent contamination.
- **Assumption 4: Two-dimensional Pauli-only noise** All noise operators stay within the qubit computational subspace and admit Pauli decomposition:

$$E = \alpha I + \beta X + \gamma Y + \delta Z$$

No population leakage outside the 2-level Hilbert space.

Four ideal assumptions form the fundamental foundation of the entire QEC framework.

Four realistic hardware noise types violating the above assumptions:

- *Level leakage propagation noise (CZ-coupling induced)*
 - Violates independence: Interaction $H_{\text{int}} \propto \hat{n}_i \hat{n}_j$ enables leakage correlation, $P(\text{leak}_i, \text{leak}_j) \gg P(\text{leak}_i)P(\text{leak}_j)$.
 - Violates sparsity: Leakage propagates via two-qubit gates, forming error clusters with weight $> t$.
 - Violates instantaneousness: Excited $|2\rangle$ states possess long relaxation time $T_1^{(2)}$; leakage persists across successive stabilizer cycles.
 - Violates two-dimensional subspace assumption: Leakage occupies higher energy subspaces; noise operators cannot be expressed as linear combinations of $\{I, X, Y, Z\}$.
- *Measurement-induced leakage noise*
 - Violates independence: Shared readout drive induces simultaneous out-of-subspace transitions on multiple ancillas.
 - Violates sparsity: Batch readout generates spatially extended synchronous leakage clusters.
 - Violates instantaneousness: Leaked population survives measurement and accumulates throughout subsequent gate layers.
 - Violates two-dimensional subspace assumption: Resonant readout pulses excite states outside $\text{span}\{|0\rangle, |1\rangle\}$.
- *Quasiparticle noise from high-energy particle impacts*
 - Violates independence: Particle strikes create spatially extended correlated faults satisfying $P(e_{\text{cluster}}) \gg \prod_{i \in \text{cluster}} P(e_i)$.
 - Violates sparsity: A single impact corrupts regions far larger than the maximum tolerable error weight t .
 - Violates instantaneousness: Quasiparticle lifetimes span numerous correction cycles, yielding persistent syndrome anomalies.

- Violates two-dimensional subspace assumption: Simultaneously generates Pauli decoherence and multi-level leakage.
- *Multi-qubit correlated Pauli noise via gate crosstalk*
 - Violates independence: Parasitic coupling yields high-probability simultaneous multi-qubit Pauli errors.
 - Violates sparsity: Coherent crosstalk generates weight- > 2 errors within a single cycle.
 - Violates instantaneousness: Static stray couplings reproduce identical multi-qubit error patterns periodically.
 - Respects two-dimensional subspace constraint: Noise stays within $\text{span}\{|0\rangle, |1\rangle\}$ but invalidates minimum-weight decoding heuristics.

Example: Leakage channel (truncated 3-level model $\{|0\rangle, |1\rangle, |2\rangle\}$)

Leakage describes population escaping the computational subspace $\text{span}\{|0\rangle, |1\rangle\}$ into higher energy levels of transmon. Consider incoherent leakage transition $|1\rangle \rightarrow |2\rangle$ with strength γ . The Kraus operators read

$$E_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{1-\gamma} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \sqrt{\gamma} & 0 \end{pmatrix},$$

satisfying $\sum_k E_k^\dagger E_k = I_3$. The quantum channel is

$$\mathcal{E}(\rho) = E_0 \rho E_0^\dagger + E_1 \rho E_1^\dagger.$$

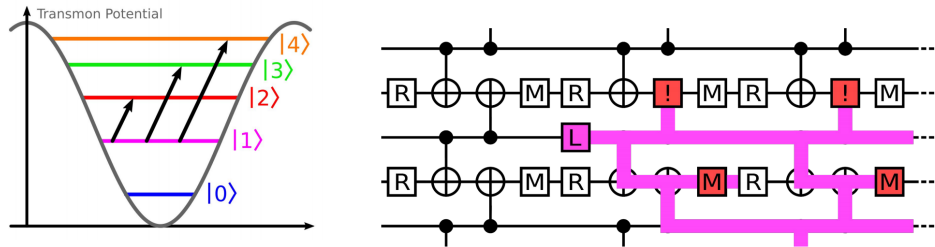
For initial state $\rho = |1\rangle\langle 1|$, we have

$$\mathcal{E}(|1\rangle\langle 1|) = (1-\gamma)|1\rangle\langle 1| + \gamma|2\rangle\langle 2|.$$

Notice that leakage **cannot be modeled by any two-level quantum channel**.

Quantum errors are more complex, eg: leakage

There is a lot more to deal with than just classical bit flips...



Assumption: We consider the pauli channel

$$\mathcal{E}(\rho) = \sum_{P \in \mathcal{P}_n} p_P P \rho P.$$

Appendix

Define partial trace by

$$\text{Tr}_B \triangleq \text{id}_{\mathcal{L}(\mathcal{H}_A)} \otimes \text{tr} : \mathcal{L}(\mathcal{H}_A) \otimes \mathcal{L}(\mathcal{H}_B) \rightarrow \mathcal{L}(\mathcal{H}_A)$$

Basis ordering convention: $|i\rangle_A = (0, \dots, 1, \dots, 0)^\top \in \mathbb{F}^{n \times 1}$, $|j\rangle_B = (0, \dots, 1, \dots, 0)^\top \in \mathbb{F}^{m \times 1}$.

The basis vector $|i\rangle_A \otimes |j\rangle_B$ corresponds to unit vector with entry 1 at position $(i-1)m + j$.

Lemma 8

$$\text{Tr}_B(\rho_{AB}) = \sum_i (\mathbb{I}_A \otimes \langle i|_B) \rho (\mathbb{I}_A \otimes |i\rangle_B), \quad \text{Tr}_A(\rho_{AB}) = \sum_i (\langle i|_A \otimes \mathbb{I}_B) \rho (|i\rangle_A \otimes \mathbb{I}_B)$$

Proof Suppose $\rho = \sum_k A_k \otimes B_k$, then

$$\begin{aligned} \text{id} \otimes \text{tr} \left(\sum_k A_k \otimes B_k \right) &= \sum_k A_k \text{tr}(B_k) = \sum_k A_k \otimes \sum_i \langle i|_B B_k |i\rangle_B \\ &= \sum_k \left[(\mathbb{I}_A A_k \mathbb{I}_A) \otimes \sum_i \langle i|_B B_k |i\rangle_B \right] \\ &= \sum_{k,i} (\mathbb{I}_A A_k \mathbb{I}_A) \otimes (\langle i|_B B_k |i\rangle_B) \\ &= \sum_{k,i} (\mathbb{I}_A \otimes \langle i|_B) (A_k \otimes B_k) (\mathbb{I}_A \otimes |i\rangle_B) \\ &= \sum_i (\mathbb{I}_A \otimes \langle i|_B) \rho (\mathbb{I}_A \otimes |i\rangle_B) \end{aligned}$$

■

Proposition 9 Let $\rho \in \mathcal{L}(\mathcal{H}_A \otimes \mathcal{H}_B)$. Decompose ρ into a block matrix labelled by indices of $\mathcal{H}_A$:

$$\rho = \begin{pmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2n} \\ \vdots & \vdots & & \vdots \\ \rho_{n1} & \rho_{n2} & \cdots & \rho_{nn} \end{pmatrix},$$

where each block satisfies $\rho_{ij} \in \mathcal{L}(\mathcal{H}_B)$. Then

- $\text{tr}_A(\rho) = \rho_{11} + \rho_{22} + \cdots + \rho_{nn}$;
- $\text{tr}_B(\rho) = \begin{pmatrix} \text{tr}(\rho_{11}) & \text{tr}(\rho_{12}) & \cdots & \text{tr}(\rho_{1n}) \\ \text{tr}(\rho_{21}) & \text{tr}(\rho_{22}) & \cdots & \text{tr}(\rho_{2n}) \\ \vdots & \vdots & & \vdots \\ \text{tr}(\rho_{n1}) & \text{tr}(\rho_{n2}) & \cdots & \text{tr}(\rho_{nn}) \end{pmatrix}.$

Proof Write $X = (X_{ij})$ in the basis of $\mathcal{H}_A$. By the matrix representation of Kronecker product,

$$X \otimes Y = \begin{pmatrix} X_{11}Y & X_{12}Y & \cdots & X_{1n}Y \\ X_{21}Y & X_{22}Y & \cdots & X_{2n}Y \\ \vdots & \vdots & & \vdots \\ X_{n1}Y & X_{n2}Y & \cdots & X_{nn}Y \end{pmatrix},$$

so the blocks satisfy $\rho_{ij} = X_{ij}Y$.

$$\sum_i \rho_{ii} = \sum_i X_{ii}Y = \text{tr}(X)Y = \text{tr}_A(X \otimes Y).$$

Moreover, the (i, j) entry of $\text{tr}_B(X \otimes Y)$ is

$$\langle i|_A \text{tr}_B(X \otimes Y)|j\rangle_A = \langle i|_A X|j\rangle_A \text{tr}(Y) = \text{tr}(X_{ij}Y) = \text{tr}(\rho_{ij}).$$

Thus both formulas hold for any Kronecker product $\rho = X \otimes Y$.

Every operator on $\mathcal{H}_A \otimes \mathcal{H}_B$ can be written as finite linear combination

$$\rho = \sum_k X_k \otimes Y_k.$$

The mapping $\rho \mapsto \text{tr}_A(\rho)$ and the block mapping are linear in ρ . Similarly for tr_B . By linearity, the identities extend to all $\rho \in \mathcal{L}(\mathcal{H}_A \otimes \mathcal{H}_B)$. ■

Proposition 10 Let U, V be vector spaces over a field $\mathbb{F}$. For $u \in U, v \in V$,

$$u \otimes v = 0 \iff u = 0 \text{ or } v = 0.$$

Proof $(\Leftarrow)$ obviously. $(\Rightarrow)$ Suppose $u \otimes v = 0$ and assume $u \neq 0$. Since $u \neq 0$, there exists a linear functional $f \in U^*$ such that $f(u) = 1$. Apply the linear map $f \otimes \text{id}_V : U \otimes V \rightarrow V$ to both sides:

$$0 = (f \otimes \text{id}_V)(u \otimes v) = f(u)v = v.$$

Hence $v = 0$. This completes the proof. ■

Proposition 11 Let $u \otimes v = u' \otimes v' \in U \otimes V$ and suppose $u \otimes v \neq 0$. Then there exists $\lambda \in \mathbb{F} \setminus \{0\}$ satisfying

$$u' = \lambda u, \quad v' = \lambda^{-1}v.$$

Proof Since $u \otimes v \neq 0$, by the previous proposition we have $u \neq 0, v \neq 0$.

First we claim u and u' are linearly dependent. Suppose for contradiction that $\{u, u'\}$ is linearly independent. Choose $f \in U^*$ with $f(u) = 1, f(u') = 0$. Applying $f \otimes \text{id}_V$:

$$v = (f \otimes \text{id}_V)(u \otimes v) = (f \otimes \text{id}_V)(u' \otimes v') = 0 \cdot v' = 0,$$

contradicting $v \neq 0$. Thus there exists some scalar $\lambda \in \mathbb{F}$ such that $u' = \lambda u$.

Substitute into the equality:

$$u \otimes v = \lambda u \otimes v' = u \otimes (\lambda v').$$

Rearranging gives $u \otimes (v - \lambda v') = 0$. As $u \neq 0$, applying Proposition 1 yields $v - \lambda v' = 0$, i.e. $v = \lambda v'$.

If $\lambda = 0$, then $u' = 0$, which implies $u' \otimes v' = 0$, contradicting $u \otimes v \neq 0$. Hence $\lambda \neq 0$, and we obtain $v' = \lambda^{-1}v$. ■

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